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Sleep Health and Lifestyles project 2
Predictive modelling

Contents

Table of Contents	i
1 Regression	1
1.1 Part A	1
1.2 Part B	3
2 Classification	4
3 Discussion and Conclusion	8
4 Appendix	9

1 Regression

1.1 Part A

Our aim in this section is to predict sleep duration in hours (y) from the independent variables (X) listed in Table 1. We chose to include the first two principal components, PC1 and PC2, from project 1 (see Figure 2 in the Appendix) and in addition, we included other attributes expected to improve the model's predictive performance.

We aim to examine how sleep duration may be affected by several factors. We used one-of-K coding for all categorical variables except occupations, which allows the model to handle non-numeric attributes. To avoid redundancy, one category was omitted; for instance, if Male=1, then female must be 0. Or if two of the columns were false, then the third must be true. Additionally, we decided to exclude the attribute 'occupation' as it contains many unique job titles that would require extensive one-of-K encoding, making the dataset unnecessarily complex and harder to interpret.

Finally, we note that in project 1, we mistakenly included the attribute 'sleep duration' in the PCA, forgetting that this is the variable we want to predict in this project. This minor mistake was corrected by redoing the PCA (as shown in appendix) resulting in nearly the same explained variance - approximately 76% for the first two components.

After defining the features and transformations, we now proceed to build a ridge regression model. In this section, we introduce the regularization parameter λ and examine how it affects the models performance. This is done using 10-fold cross-validation, where the model is trained and validated across different values of λ in a logarithmic scale $\lambda \in \{-3, 3, 30\}$. From this analysis we determine the optimal λ (see Figure 1), which is 3.29. The difference between the training and validation errors is very small, indicating that the model generalizes well and shows no signs of overfitting. Both errors are also low (approximately 0.100), suggesting that the model explains a large part of the variation in the dependent variable, sleep duration.

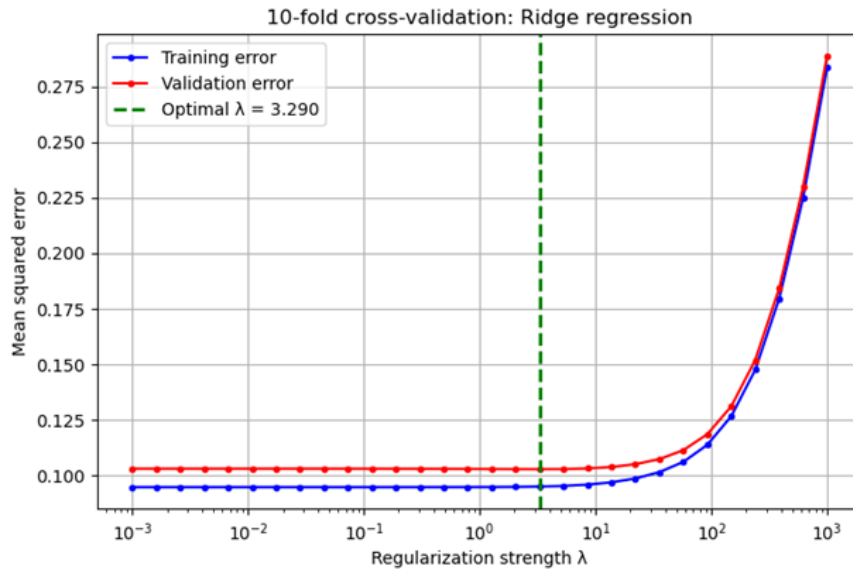


Figure 1

After computing the ridge regression, we obtain the weights as shown in the table below (Table 1). These weights indicate how each independent variable contributes to predicting sleep duration. A positive weight means that the variable increases sleep duration, while a negative weight decreases it.

We can see that quality of sleep has the strongest positive contribution, suggesting that people with better sleep quality tend to sleep longer. Stress level has a negative weight, which means that higher stress is associated with shorter sleep duration. Other variables, such as PCA2 and blood pressure (prehigh) also have smaller positive effects, while BMI (overweight) shows a negative contribution.

Table 1: Ridge regression coefficients with $\lambda = 3.290$

Feature	Coefficient
PCA1	0.0583
PCA2	0.0768
Quality of Sleep	0.4655
Stress Level	-0.3108
Gender (Male)	0.0349
BMI Category (Obese)	-0.0147
BMI Category (Overweight)	-0.1415
Blood Pressure (Pre-high)	0.2288
Blood Pressure (Stage 1 high)	0.1682
Sleep Disorder (No disorder)	0.0108
Sleep Disorder (Sleep Apnea)	0.0536

1.2 Part B

In this section, we compare three models to evaluate their ability to predict Sleep Duration: the regularized linear regression model (from Part A), an artificial neural network (ANN), and a baseline model. The baseline predicts the mean sleep duration from the training set, providing a minimal reference level of performance.

A two layered 10 x 10 cross validation is the used method. The outer loop estimates the generalization error of each model, while the inner loop performs hyperparameter selection. For Ridge regression, the regularization parameter λ is tuned on a logarithmic scale identical to the one in Part A. For the ANN, we tested hidden layers sizes $h \in \{1, 2, 4, 8, 16\}$ to balance model complexity and computational efficiency. Both models minimize the Mean Squared Error (MSE) on the training data. The baseline model was evaluated only in the outer cross-validation loop, as it does not involve any hyperparameter tuning. The performance of all models across the outer folds is summarized in the following table, highlighting each fold's optimal hyperparameters and corresponding test errors (Table 2).

Table 2: Two-level cross-validation table used to compare the three models

Outer fold	ANN		Rigde regression		Baseline
	h_i^*	E_i^{test}	λ_i^*	E_i^{test}	
1	16	0.0713	3.290	0.1044	0.5294
2	16	0.0379	8.532	0.0961	0.5985
3	16	0.0678	3.290	0.1099	0.6619
4	16	0.0491	2.043	0.0865	0.7061
5	8	0.1090	3.290	0.1000	0.7371
6	16	0.0626	5.298	0.0981	0.6158
7	16	0.0617	5.298	0.1236	0.6853
8	16	0.1398	3.290	0.1049	0.5221
9	8	0.0642	5.298	0.1099	0.6251
10	16	0.0733	3.290	0.0859	0.6639
Mean		0.0737		0.1019	0.6345

The MSE results clearly show that the baseline model underperforms compared to the two other learning models, as it has a substantially higher test error. On average, the ANN achieved the lowest test error (0.0737), followed closely by Ridge Regression (0.1019). However, there is some variation across folds. In a few cases, Ridge Regression slightly outperformed the ANN, indicating that Ridge was a close competitor.

These variations raise the question of whether the observed performance differences between the models are statistically significant or could have occurred by chance. We therefore used method 11.4.1, where we conducted a paired correlated t-test using a significance level $\alpha = 0.05$. The results is presented in the following table (Table 3).

Table 3: Pairwise statistical evaluation models

Model A	Model B	Mean diff	95% CI	t-stat	p-value
Ridge	Baseline	-0.5326	[-0.5849, -0.4803]	-23.0412	0.0000
ANN	Baseline	-0.5609	[-0.6203, -0.5014]	-21.3411	0.0000
ANN	Ridge	-0.0283	[-0.0500, -0.0065]	-2.9426	0.0164

Based on the p-values, all model comparisons show statistically significant differences ($p < 0.05$). This allows us to reject the null hypothesis that the models perform equally. In particular, both Ridge Regression and the ANN significantly outperform the baseline model. The difference between ANN and Ridge Regression is also statistically significant, though smaller in scale. While the t-test is not an absolute measure of model comparison, it provides some evidence that the ANN's outperformance is unlikely to be due to random variation.

2 Classification

As mentioned in the first report, for the classification problem, we have chosen to predict whether an individual has a sleep disorder based on the other available attributes in the dataset. The target variable is **Sleep Disorder**, which includes three categories: No Disorder, Insomnia, and Sleep Apnea. Since the target variable has more than two possible classes, this is a multi-class classification problem.

To begin the analysis, we first separate the data into features (X) and the target (y):

- The target y is set to Sleep Disorder.
- The features X include all other attributes.

However, not all columns are useful for prediction. Some variables are misleading and can negatively impact the model's performance. Therefore, we chose to work with the attributes mentioned in Table 1.

In this section, we compare three different models: **Logistic Regression**, **K-Nearest Neighbors (KNN)**, and a **Baseline** model. Each model represents a distinct approach to classification.

For the **Logistic Regression** model, we use a multinomial logistic regression approach that can handle more than two classes. It predicts the probability that each person belongs to one of the three sleep disorder categories. To prevent overfitting, we add a regularization term that penalizes large weights in the model. This penalty is controlled by the parameter λ , which adjusts how strong the regularization is. Smaller values of λ make the model more flexible (less regularization), while larger values make it simpler and reduce overfitting. After a few test runs, we decided to search across a logarithmic range of values:

$$\lambda \in [10^{-12}, 10^2]$$

In the **KNN** model, the main parameter that controls complexity is k , the number of neighbors. A small k makes the model very flexible and more likely to follow noise in the data, while a large k makes it smoother but less sensitive to small changes. To find a good balance, we tested different values of k in the range:

$$k \in [1, 10]$$

The **Baseline** model is used as a simple comparison. It always predicts the most common class in the training data, no matter what the input features are. This gives us a reference point to check whether our other models are actually learning meaningful patterns.

Using the two-level cross-validation approach, we were able to evaluate the performance of our models. In this setup, the data is split into two loops:

- The inner loop is used to find the best parameters for each model. For Logistic Regression, this means finding the best value of λ , and for KNN, the best value of k .
- The outer loop is used to test how well the model generalizes to unseen data. Once the best parameters are chosen in the inner loop, we train the model with those settings on the outer training set and test it on the outer test set.

The performance is measured using the error rate, which is the fraction of samples that the model classifies incorrectly:

$$E = \frac{\text{Number of misclassified observations}}{N^{\text{test}}}$$

The error rate for each fold and model is shown in Table 4. The table also includes the best hyperparameters found in each outer fold.

Table 4: Classification models compared through two-level cross-validation

Outer fold	KNN		Logistic Regression		Baseline
i	k_i^*	E_i^{test}	λ_i^*	E_i^{test}	E_i^{test}
1	3	0.105	3.16e-05	0.132	0.421
2	5	0.105	1.00e-05	0.105	0.421
3	8	0.184	3.16e-07	0.132	0.421
4	5	0.105	1.00e-05	0.079	0.421
5	5	0.081	3.16e-06	0.054	0.405
6	5	0.054	4.33e-09	0.000	0.405
7	7	0.054	3.16e-07	0.135	0.405
8	5	0.189	3.16e-04	0.135	0.405
9	5	0.108	3.59e-03	0.135	0.405
10	5	0.162	3.16e-06	0.162	0.432
Mean	–	0.115	–	0.107	0.414

From these results, both Logistic Regression and KNN perform much better than the baseline, which simply predicts the most common class. The Logistic Regression model achieved the lowest average test error (about 10.7%), which corresponds to an accuracy of roughly 89.3%. This means it made the fewest misclassifications overall. The KNN model had a slightly higher error rate (around 11.5%), but still performed well. The baseline model performed significantly worse, with a test error of over 41%, showing that both of our trained models learned useful patterns in the data.

Overall, the two-level cross-validation shows that both the Logistic Regression and KNN models generalize well to unseen data. Logistic Regression performs slightly better, but the difference between the two models is small, which suggests that both are suitable for predicting sleep disorders.

To compare the performance of the three models, we performed a pairwise statistical evaluation using the **McNemar test**. It focuses on the number of instances where the models disagree, making it ideal for paired comparison. For each comparison, we calculated the difference in proportions of misclassified samples, denoted as θ , along with a 95% confidence interval and the corresponding p -value. The results are shown below.

Table 5: Pairwise statistical evaluation

Model A	Model B	θ	95% CI Low	95% CI High	p-value
KNN	Logistic Regression	0.0080	-0.0168	0.0302	0.6476
KNN	Baseline	-0.2995	-0.3299	-0.2552	<0.001
Logistic Regression	Baseline	-0.3075	-0.3431	-0.2588	<0.001

From Table 5, the p -value between KNN and Logistic Regression (0.65) indicates that there is no significant difference between these two models. They perform almost equally well. However, both KNN and Logistic Regression show highly significant differences when compared with the Baseline model ($p < 0.001$). This means that both learning models perform much better than the baseline, which simply predicts the majority class.

The confidence intervals tell us whether the differences between models are meaningful or just due to chance. For KNN and Logistic Regression, the 95% confidence interval includes zero, which means their performances are statistically similar. However, the intervals for both models compared to the baseline do not include zero, showing that KNN and Logistic Regression perform significantly better than the baseline model.

Based on the statistical results, both Logistic Regression and KNN significantly outperform the Baseline model. Between these two, Logistic Regression achieved slightly lower error rates and a higher mean accuracy, but this difference is not statistically significant. Therefore, both models can be considered equally effective for predicting sleep disorders, though Logistic Regression is preferred due to its simplicity.

The logistic regression model makes predictions by calculating the probability that each observation belongs to one of the three sleep disorder classes. It does this using a linear combination of the input features, where larger positive coefficients increase the likelihood of belonging to a specific class. The class with the highest predicted probability is chosen as the model's output.

When examining the model's coefficients, we found that the most influential features were similar to those identified in the regression part of the report, even though they predict two different variables. In particular, *Quality of Sleep*, *Blood Pressure*, and *Stress Level* remained among the most important predictors of sleep disorders and sleep duration. In summary, the logistic regression model uses the same key features as in the regression analysis, confirming that these variables play a consistent role in understanding sleep health.

3 Discussion and Conclusion

In this project, we developed models to predict a person's sleep duration and sleeping disorder based on various influencing factors. In the ridge regression model, we found the optimal regularization parameter λ to be quite low. Furthermore, there was no significant increase in the validation error as λ decreased, suggesting that the model does not benefit much from regularization. This indicates that the dataset is well-structured and does not suffer from issues related to bias or high variance.

For the comparison of the three prediction models (Ridge, ANN and Baseline) Baseline was the worst performing, as expected, since it only predicts the mean sleep duration and does not account for any input features. Although the ridge regression model performed well with a low error rate, the Artificial Neural Network (ANN) achieved better performance overall. When observing the optimal h for the ANN, we noticed a tendency toward higher values, which gives the model more capacity to represent nonlinear patterns. This suggests that the underlying relationships in the data are at least partially nonlinear, which explains the ANN's higher accuracy. Therefore, if this work were to be continued, it could be beneficial to explore other nonlinear method.

In the classification part of this project, we aimed to predict whether an individual suffers from a sleep disorder based on different lifestyle and health-related attributes. Three models were compared: Logistic Regression, K-Nearest Neighbors (KNN), and a Baseline model.

The results from the two-level cross-validation showed that both Logistic Regression and KNN performed much better than the Baseline model. Logistic Regression achieved the lowest average test error, followed closely by KNN. This indicates that both models were able to learn useful patterns from the data, with Logistic Regression showing slightly better generalization.

The McNemar test confirmed that the performance difference between Logistic Regression and KNN was not statistically significant. However, both models were found to be significantly better than the Baseline. This demonstrates that both approaches provide meaningful predictive power for identifying sleep disorders.

Overall, the classification results show that simple models such as Logistic Regression and KNN can effectively classify sleep disorders based on lifestyle and health features. Logistic Regression was slightly preferred due to its interpretability and stability across validation folds.

No previous studies were found that have applied regression or classification methods to this dataset. Therefore, no comparison could be made.

4 Appendix

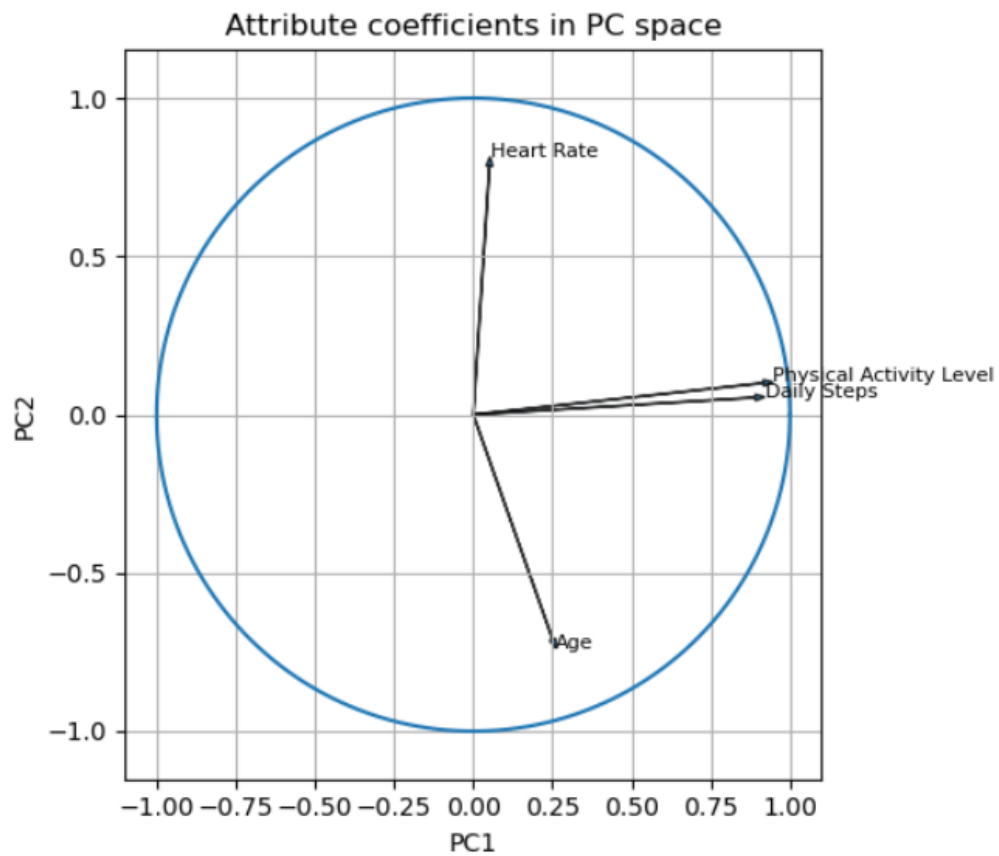


Figure 2

References

- [1] OpenAI. *ChatGPT*. ChatGPT has been used to formulate sentences in the report. 2025. URL: <https://chat.openai.com/>.